	GLOBAL ECONOMICS Grade: 10TH Learning Evidence 1 Teacher: Nicolás López Cuéllar	1st TERM 2026–2027

Learning objective: Structure decision-making problems under uncertainty, calculate and organise payoffs, and apply the Maximax and Maximin criteria to compare alternatives and justify decisions.	**Assessment criteria** C1. Characterises decision-making problems under uncertainty by identifying decision alternatives, states of nature, consequences, and the economic or financial context of the decision. C2. Organises decision problems using payoff tables, placing alternatives, states of nature, and payoffs in a clear structure for comparison. C3. Calculates profits by combining revenues with fixed and variable costs across different states of nature, determining the corresponding payoffs for each alternative, and organising the results in payoff tables for comparison. C4. Applies the Maximin and Maximax criteria to make optimistic or conservative decisions without probabilities, selecting the alternative with the highest possible payoff or the best worst-case, interpreting risks or limitations.

Student's name: ____________________ **Group:** ________ **Date:** ________

Criteria assessment

This task sheet assesses criteria C1 through C4. Each problem assesses the criteria indicated in its title. The number of points assigned to each task is shown at the end of the item. Partial credit may be awarded when the correct procedure is shown but arithmetic errors are made. Answers that do not show the required procedure will not receive credit. A minimum of 4 points is required to pass each criterion.

1. Local Food Market Stall [C1-1 C2-2]

A local food market company is planning its stall offering for an upcoming market period. A standard stall is expected to generate profits of eighteen thousand US dollars under high demand and eleven thousand US dollars under low demand, while a deluxe stall is expected to generate thirty-one thousand US dollars and six thousand US dollars, respectively. The company must select its stall before the actual level of customer demand is known. Characterize the key elements of the decision-making situation and construct the payoff table.

2. Small Bakery Production [C1-1 C2-3]

A small bakery is planning its production capacity for an upcoming sales period. A standard oven can generate 3 thousand BRL per production batch, while an industrial oven can generate 5 thousand BRL per batch. Based on expected demand, the standard oven would complete 16 batches under high demand or 11 under low demand, whereas the industrial oven would complete 13 or 9 batches, respectively. In addition, the bakery would receive a fixed sales bonus of 6 thousand BRL if demand is high and incur a fixed loss of 3 thousand BRL if demand is low. Characterize the key elements of the decision-making situation and construct the payoff table.

3. Urban Meal Subscription [C1-2 C2-5]

An urban meal subscription company must choose between a basic meal plan and a premium meal plan before customer interest is known. Interest may be strong or weak. The basic plan generates 4 thousand BRL per active subscriber and also provides a fixed operating contribution of 6 thousand BRL. The premium plan generates 6 thousand BRL per active subscriber and provides a fixed operating contribution of 9 thousand BRL. Strong interest adds another 2 thousand BRL per active subscriber to either plan, while weak interest adds 1 thousand BRL per active subscriber. The basic plan is expected to have 13 active subscribers under strong interest and 8 under weak interest; the premium plan is expected to have 10 active subscribers under strong interest and 6 under weak interest. Characterize the key elements of the decision-making situation and construct the payoff table.

4. Regional Bookstore Expansion [C1-1 C3-2]

A regional bookstore must decide whether to open a children's section or maintain its current product selection before customer traffic is known. Traffic may be strong or weak. All values are in million BRL. Opening the children's section produces revenue of 72 under strong traffic and 35 under weak traffic, while maintaining the current selection produces revenue of 61 under strong traffic and 47 under weak traffic. The corresponding costs are 24 and 22 for the children's section under strong and weak traffic, respectively, and 25 and 23 for the current selection under strong and weak traffic, respectively. Characterize the key elements of the decision-making situation and construct the payoff table.

5. Greenhouse Crop Production [C1-1 C3-3]

A greenhouse farm is planning its production for an upcoming growing season. Revenue per production lot is 18 thousand EUR for tomatoes and 15 thousand EUR for peppers. Under favourable weather conditions, the farm expects to produce 8 lots of tomatoes and 7 lots of peppers, while under unfavourable weather conditions it expects to produce 5 lots of tomatoes and 4 lots of peppers. The cost per production lot is 7 thousand EUR under favourable weather and 10 thousand EUR under unfavourable weather. Characterize the key elements of the decision-making situation and construct the payoff table.

6. Fitness Center Membership [C1-2 C3-5]

A fitness center must choose between a basic membership plan and a premium membership plan before monthly demand is known. Demand may be high or low. Quantities, in hundreds of memberships, are 9 and 6 for the basic plan and 7 and 4 for the premium plan under high and low demand, respectively. The plan-based revenue rates are 8 and 13 thousand USD per hundred memberships for the basic and premium plans. High demand adds a state-based revenue rate of 2 thousand USD per hundred memberships; low demand adds 1. The plan-based cost rates are 3 and 6 thousand USD per hundred memberships, respectively. The fixed operating costs are 12 thousand USD under high demand and 16 thousand USD under low demand. Characterize the key elements of the decision-making situation and construct the payoff table.

7. Restaurant Expansion Strategy [C4-2]

A restaurant company must choose a strategy for expanding into a new location. Customer demand could be high, medium, or low. All payoffs are in thousands of dollars.

	Demand		
Alternative	High	Medium	Low
Full-service restaurant	78	35	8
Casual restaurant	72	48	32
Express restaurant	95	42	28
Postpone expansion	25	25	25

What would the optimistic decision be? What would the conservative decision be? Justify your answer.

8. Hotel Development Strategy [C4-2]

A hotel company must choose a strategy for developing a new property. Tourist demand could be high, medium, or low. All payoffs are in thousands of dollars.

	Tourist demand		
Alternative	High	Medium	Low
Large hotel	72	34	10
Medium hotel	65	46	22
Small hotel	95	44	31
No development	58	40	35

What would the optimistic decision be? What would the conservative decision be? Justify your answer.

9. Agricultural Technology Investment [C4-2]

An agricultural company must choose a strategy for introducing a new automated farming system. Market conditions could be highly favourable, favourable, stable, or unfavourable. All payoffs are in thousands of dollars.

S1 Highly favourable market conditions.
S2 Favourable market conditions.
S3 Stable market conditions.
S4 Unfavourable market conditions.

	S1	S2	S3	S4
Full automation	125	58	8	-42
Partial automation	88	64	30	4
Basic automation	62	52	38	20
Equipment leasing	48	43	35	28

What would the optimistic decision be? What would the conservative decision be? Justify your answer.